

8. Even More Integral Applications

8.5. Probability

8.5 Learning Objectives

- Understand how we describe probability of events using integrals
- Identify whether a function can be a probability density function
- Find probability of events occurring using a PDF
- Find the mean value of a PDF

Definition 8.5.1 Probability

The **probability** of an event is how likely the event is to occur, and is restricted to values between $[0, 1]$.

Example 8.5.1

If you flip a coin, the probability it lands on either side is 0.5 (which we normally translate to 50%).

Definition 8.5.2 Discrete Random Variable

X is a **discrete random variable** if it is a number that represents an outcome that can only be certain distinct values.

Example 8.5.2

The result of rolling a six sided die, or the number of times you need to flip a coin before getting “heads” are two examples of discrete random variables. The first has a finite set of outcomes: $\{1, 2, 3, 4, 5, 6\}$, the second has a countable set of outcomes: $\{1, 2, 3, 4, 5, 6, 7, 8, \dots\}$ (depending on how unlucky you are).

8.5.1. Continuous Random Variables and Probability Density

We'll spend our time in probability land here:

Definition 8.5.3 Continuous Random Variable

X is a **continuous random variable** if it is a number that represents an outcome that can take any real value on some interval.

Example 8.5.3

- The height of a randomly selected person on Earth is a continuous random variable, the person *could* be any height from 0 feet to 10 feet tall, so the random variable has a set of outcomes $[0, 10]$. (In a fancy probability class you'd call this interval the support of the random variable, but you don't need to know that for now)
- The number of inches a die rolls after dropping it is a continuous random variable, the die *could* roll anywhere from 0 inches to 12 inches (because your dice rolling tray is only 12 inches long and has negligible width ;)). This random variable has a set of outcomes in $[0, 12]$.
- The number of hours before the Colorado Rockies win a Baseball game is a continuous random variable (so long as we are being VERY exact about our timing (t can be any positive real number) and taking care to think about when the game is, etc). The Rockies *could* win their next game (we'll assume it ends in 2 hours) or they may never win again, so this random variable can have an outcome in $[120, \infty)$.

Property 8.5.1 Probability Density Function

Every continuous random variable, X has a **probability density function**, $f(x)$, such that

- The probability that X is between a and b is given by

$$P(a \leq X \leq b) = \int_a^b f(x) dx.$$

- The probability density function is always non-negative,

$$f(x) \geq 0$$

- The “total probability” is 1, that is

$$\int_{-\infty}^{\infty} f(x) dx = 1$$

Note

For a continuous random variable X , the probability that $X = c$ for any real c in the set of possible outcomes of X is 0. This is because no matter the probability density function $f(x)$, $\int_c^c f(x) dx = 0$.

Fun Fact

Although you do not have the tools to integrate e^{-x^2} , a scaled version of this is a probability density function that represents the normal distribution! It has domain $(-\infty, \infty)$ Specifically,

$$f(x) = \frac{1}{\sqrt{2\pi}\sigma^2} \exp\left(-\frac{(x - \mu)^2}{2\sigma^2}\right)$$

for some constants μ, σ . So, we know that

$$\int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right) dx = 1.$$

Example 8.5.4

Let

$$f(x) = \begin{cases} \frac{1}{6}x(10-x) & 0 \leq x \leq 10 \\ 0 & \text{otherwise} \end{cases}$$

Could $f(x)$ be a PDF for some continuous random variable?

Solution

We need to check 2 things: Is $f(x) \geq 0$, and is $\int_{-\infty}^{\infty} f(x) dx = 1$?

First, for $0 \leq x \leq 10$, we have $\frac{1}{6}x \geq 0$ and $10 - x \geq 0$ and the product of 2 non-negative numbers is non-negative. Outside this interval $f(x) = 0 \geq 0$ so yes, $f(x) \geq 0$ everywhere.

Second, we check the integral

$$\begin{aligned} \int_{-\infty}^{\infty} f(x) dx &= \int_{-\infty}^0 0 dx + \int_0^{10} \frac{1}{6}x(10-x) dx + \int_{10}^{\infty} 0 dx \\ &= 0 + \int_0^{10} \frac{1}{6}x(10-x) dx + 0 \\ &= \frac{1}{6} \int_0^{10} 10x - x^2 dx \\ &= \frac{1}{6} \left[5x^2 - \frac{x^3}{3} \right] \Big|_0^{10} \\ &= \frac{1}{6} \left(500 - \frac{1000}{3} \right) \\ &= \frac{500}{18} \neq 1. \end{aligned}$$

So, $f(x)$ could not be a PDF for some continuous random variable.

Exercise 8.5.1

Show that $f(x)$ is a PDF for some random variable X .

$$f(x) = \begin{cases} \frac{\pi}{2} \sin(\pi x) & 0 \leq x \leq 1 \\ 0 & \text{otherwise} \end{cases}$$

Example 8.5.5

Find $P(X \geq \frac{1}{2})$, if the PDF for X is

$$f(x) = \begin{cases} \frac{\pi}{2} \sin(\pi x) & 0 \leq x \leq 1 \\ 0 & \text{otherwise} \end{cases}$$

Solution

We can assume this is a PDF (because we showed it in previous exercise, but also the wording of the problem). Then all we do is find

$$\begin{aligned} P\left(X \geq \frac{1}{2}\right) &= \int_{\frac{1}{2}}^{\infty} f(x) \, dx \\ &= \int_{\frac{1}{2}}^1 \frac{\pi}{2} \sin(\pi x) \, dx + \int_1^{\infty} 0 \, dx \\ &= \frac{\pi}{2} \int_{\frac{1}{2}}^1 \sin(\pi x) \, dx \\ &= \frac{\pi}{2} \frac{1}{\pi} \int_{\frac{\pi}{2}}^{\pi} \sin(u) \, du \\ &= \frac{1}{2} (-\cos(u)) \Big|_{\frac{\pi}{2}}^{\pi} \\ &= \frac{1}{2} \left(\cos\left(\frac{\pi}{2}\right) - \cos(\pi) \right) \\ &= \frac{1}{2} (0 - (-1)) \\ &= \frac{1}{2} \end{aligned}$$

Where $u = \pi x$, $du = \pi \, dx$, $u(\frac{1}{2}) = \frac{\pi}{2}$, $u(1) = \pi$.

Note

Note, probability is always between 0 and 1. This is a good thing to double check at the end, is the probability between 0 and 1?

8.5.2. Mean Value of a Probability Density Function

Definition 8.5.4 Mean of a random variable

The average, or **mean**, or **expected value** of a continuous random variable X with probability density function $f(x)$ is

$$\mu = \int_{-\infty}^{\infty} f(x)x \, dx$$

Remark

Consider the PDF $f(x)$ on $(-\infty, \infty)$. Now, consider the lamina defined as the area under $f(x)$. What is the x coordinate of the centroid of this lamina?

$$\bar{x} = \frac{\int_{-\infty}^{\infty} f(x)x \, dx}{\int_{-\infty}^{\infty} f(x) \, dx}$$

However, since $f(x)$ is a PDF, $\int_{-\infty}^{\infty} f(x) \, dx = 1$. So,

$$\bar{x} = \frac{\int_{-\infty}^{\infty} f(x)x \, dx}{1} = \int_{-\infty}^{\infty} f(x)x \, dx.$$

How does this relate to the average of a continuous random variable with f as its PDF?

It is the same!

8.5.3. Median Value of a Probability Density Function

Definition 8.5.5 Median of a random variable

The **median** of a random variable X with probability density function $f(x)$ is M , where M is found by solving

$$\int_{-\infty}^M f(x) \, dx = \frac{1}{2}.$$

Exercise 8.5.2

Find the median of a continuous random variable X with probability density function defined as

$$f(x) = \begin{cases} \frac{\pi}{2} \sin(\pi x) & 0 \leq x \leq 1 \\ 0 & \text{otherwise} \end{cases}$$

Hint: we saw this function earlier

Example 8.5.6

While taking a walk along the road where you live, you accidentally drop your glove, but you don't know where. The probability density function $f(x)$ for having dropped your glove x kilometers from home is

$$f(x) = 2e^{-2x}, x \geq 0$$

(it is 0 otherwise)

1. What is the probability that you dropped your glove within 1 km of your home?
2. At what distance y from home is the probability that you dropped it within y km of home at least 95%?
3. What is the mean value for when you dropped your glove?
4. What is the median value for when you dropped your glove?

 Solution

1. What is the probability that you dropped your glove within 1 km of your home?

$$\begin{aligned}P(0 \leq X \leq 1) &= \int_0^1 2e^{-2x} dx \\&= \int_0^{-2} -e^u du \\&= -e^u \Big|_0^{-2} \\&= -e^{-2} - (-e^0) \\&= 1 - e^{-2} \\&\approx 0.86\end{aligned}$$

The probability the glove was dropped within km is 86%.

2. At what distance y from home is the probability that you dropped it within y km of home at least 95%?

We need to solve when $P(0 \leq x \leq y) = 0.95$, so

$$\begin{aligned}0.95 &= P(0 \leq x \leq y) \\&= \int_0^y 2e^{-2x} dx \\&= \int_0^{-2y} -e^u du \\&= \int_0^{-2y} -e^u du \\&= 1 - e^{-2y} \\&\Rightarrow -0.05 = -e^{-2y} \\&\Rightarrow 0.05 = e^{-2y} \\&\Rightarrow \ln(0.05) = -2y \\&\Rightarrow \frac{\ln(0.05)}{-2} = y\end{aligned}$$

So, at $y \approx 1.5$ km from home, there is a 95% chance you dropped your glove before this point.

3. What is the mean value for when you dropped your glove?

$$\begin{aligned}\mu &= \int_{-\infty}^{\infty} f(x)x dx \\&= \int_{-\infty}^0 0 dx + \int_0^{\infty} 2e^{-2x}x dx \\&= 0 + \int_0^{\infty} 2e^{-2x}x dx \\&= \lim_{b \rightarrow \infty} \int_0^b 2e^{-2x}x dx\end{aligned}$$

which needs integration by parts

$$u = x, \quad dv = 2e^{-2x} dx,$$

$$du = dx, \quad v = \int 2e^{-2x} dx = -e^{-2x}$$

So

$$\begin{aligned}\mu &= \lim_{b \rightarrow \infty} \left[-xe^{-2x} \Big|_0^b - \int_0^b -e^{-2x} dx \right] \\ &= \lim_{b \rightarrow \infty} \left[-\frac{x}{e^{2x}} \Big|_0^b + \int_0^b e^{-2x} dx \right] \\ &= \lim_{b \rightarrow \infty} \left[-\frac{x}{e^{2x}} \Big|_0^b + \frac{e^{-2x}}{-2} \Big|_0^b \right] \\ &= \lim_{b \rightarrow \infty} \left[-\frac{b}{e^{2b}} + \frac{0}{e^0} + \frac{e^{-2b}}{-2} - \frac{e^0}{-2} \right] \\ &= \lim_{b \rightarrow \infty} \left[-\frac{b}{e^{2b}} + \frac{e^{-2b}}{-2} + \frac{1}{2} \right] \\ &= \lim_{b \rightarrow \infty} -\frac{b}{e^{2b}} + \lim_{b \rightarrow \infty} \frac{e^{-2b}}{-2} + \frac{1}{2} \\ &= \lim_{b \rightarrow \infty} -\frac{b}{e^{2b}} + 0 + \frac{1}{2}\end{aligned}$$

This limit has indeterminate form $-\frac{\infty}{\infty}$ so we need l'Hospital's rule

$$\mu \stackrel{\text{LH}}{=} \lim_{b \rightarrow \infty} -\frac{1}{2e^{2b}} + \frac{1}{2} = 0 + \frac{1}{2} = \frac{1}{2}.$$

The mean distance for where you dropped your glove is $\frac{1}{2}$ km.

4. What is the median value for when you dropped your glove? We must solve for M in

$$\begin{aligned}
\int_{-\infty}^M f(x) \, dx &= \frac{1}{2} \\
\int_{-\infty}^0 0 \, dx + \int_0^M 2e^{-2x} \, dx &= \frac{1}{2} \\
0 + \int_0^M 2e^{-2x} &= \frac{1}{2} \\
-e^{-2x} \Big|_0^M &= \frac{1}{2} \\
1 - e^{-2M} &= \frac{1}{2} \\
e^{-2M} &= \frac{1}{2} \\
-2M &= \ln\left(\frac{1}{2}\right) \\
2M &= \ln(2) \\
M &= \frac{1}{2} \ln(2)
\end{aligned}$$

So the median distance for where you dropped your glove is $\mu = \frac{1}{2} \ln(2) \approx 0.34\text{km}$.

8.5 Section Summary:

- We discussed continuous random variables, their probability density functions, and their mean and median values.